Pedestrian Evacuation Analyst Tool User Guide

By Jeanne M Jones, U.S. Geological Survey

U.S. Department of the Interior
U.S. Geological Survey

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Overview

The Pedestrian Evacuation Analyst Tool (PEAT) software was developed to aid researchers and practitioners studying pedestrian, self-initiated evacuation from sudden-onset hazards, such as local tsunamis, debris flows, lahars (volcanic mudflows), and flash floods [1,2]. Detailed analysis of evacuation potential requires a large amount of high-resolution data and multiple processing steps. The software streamlines this process by managing the data files and providing an orderly path through the processing. Multiple geoprocessing operations are automated into as few steps as possible to free up the user's time for further analysis.

This guide provides installation instructions for the Pedestrian Evacuation Analyst toolbox into Esri's ArcGIS Pro and describes the steps to run the software and produce evacuation-time maps, speed maps, and evacuation flow lines. Publications describing the methodology and various applications of anisotropic path distance modeling are listed in the References.

Tool Disclaimer

The goal is to provide a research tool for exploratory analysis of evacuation potential on the landscape. Results are intended to initiate risk-reduction and preparedness discussions within communities and not serve as definitive statements on the potential mortality from a certain hazard scenario. First, the Least Cost Distance modeling underlying this tool is based on optimal routes across open landscapes that are determined by mathematical relationships relating slope, land cover, and movement. Actual evacuation pathways taken by at-risk individuals may vary substantially from modeled routes. Second, modeling assumes certain travel speeds that may or may not be accurate given the characteristics of the individual or their surroundings (for example, evacuations during a morning rainstorm versus one during a cloudless night sky). Third, modeling travel times to safe zones is based on specific hazard scenarios; therefore, actual safe zones during an event may be different than the model-produced safe zones based on specific scenarios or assumptions.

System Requirements

- Esri ArcGIS Pro with an Advanced license running in a Microsoft Windows environment, which should provide access to the Spatial Analyst toolbox

- Esri has updated the Distance Accumulation tool which PEAT uses for evacuation calculations. The tool version in Pro 3.0 produced reasonable results, but an algorithm change in Pro 3.1 caused incorrect modeling outputs. A corrected algorithm was released with 3.1.3 that again produced reasonable results but with slightly longer travel times.

- The time required to run the Distance Accumulation tool has also changed with different versions. In version 3.0, the tool took about 2 minutes to run for an elevation file approximately 100 Mbytes in size. The processing in 3.1.3, with the algorithm change, takes about 40 minutes for the same raster with full landscape processing. The roads-only processing takes significantly less time.

Tool Data Storage

The tool uses a file geodatabase to store feature classes that are generated during processing and uses a separate folder for raster storage. Testing showed that tool operation was more robust when the rasters were stored outside of the geodatabase. The tool uses folders and feature datasets or organize the data and make files easier to find. See the section on file storage for an illustration.

The Evacuation Analyst creates multiple output rasters during tool processing. The number of rasters depends on the number of tools run and the tool options, such as the number of vertical evacuation structures to process. Large input digital elevation models (DEMs) will create large output rasters, and the user should be mindful of available disk space when running the tool. Also, the PEAT tools are performing rapid geospatial processing with potentially large datasets. It is recommended to store tool data locally on the same machine with ArcPro. Remote or cloud datastores may cause occasional processing errors due to slower connections.

Installation

1. Verify that your ArcPro version is 3.0.x or 3.1.3 and higher. Versions of 2.9.x may work. Versions 3.1.0, 3.1.1, and 3.1.2 won't produce acceptable results. See System Requirements.

2. Create a new folder in your work area, move the zip file of the tool into it, and unzip it. This contains the toolbox and the code needed to run the tools.

3. Create another new folder in your work area and create an empty file geodatabase (gdb) in this folder. This gdb will hold your PEAT project files so name it something appropriate (like your study area). The folder for raster storage will be created automatically by the tool based on the location of the file gdb.

4. Open ArcPro and open the Catalog pane. Under the Project tab in the Catalog window, add the PEAT_tools toolbox to the project toolboxes by right-clicking Toolboxes and selecting 'Add Toolbox'. Then navigate to the PEAT_tools.tbx file. See Figure 1 for an example of the toolbox.

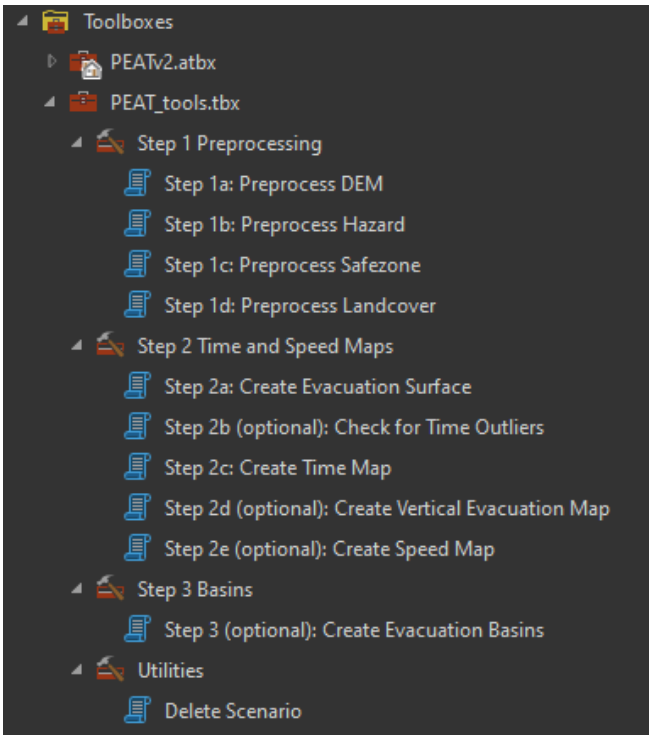


Figure 1. The contents of the PEAT_tools toolbox in ArcPro's Catalog window.

Updating the toolbox

When updating the toolbox to a newer version:

1. Delete the entire toolbox folder, including the code and any sub-folders
2. Unzip the updated toolbox folder into a new folder
3. Close ArcPro and open it again, selecting the updated toolbox
4. The toolbox does not store any references to the projects, so updating the toolbox should not require any change to your project data storage.

Tips

- The project file geodatabase will fill up with a lot of files if you run all the processing with multiple travel speeds. Check the naming convention at the end of this guide if you're not sure which file was generated during any step in the processing.
- Open the View Details window while or after running each tool to see detailed messages about the processing and completion.
- The tool descriptions below will tell you which tools will automatically load a tool result into the Table of Contents (TOC) pane. The result will be added to the active map in ArcPro, which is the map pane that is currently selected and open for view. If, during tool processing, you switch to a different map in Pro (perhaps a different study area), the tool will load the result into this active map. Likewise, if you have the Catalog View

open in the main Pro window, there is no active map and the result layer won't be added to the TOC. You can still add layers manually from the project geodatabase or raster folder by dragging them into the map pane.

Step 1 Preprocessing

These tools are used to prepare the elevation raster, safe zone feature class and raster, and cost inverse raster (created during land cover preprocessing) in the project database. These files are used in later tools to run the evacuation modeling.

Step 1a: Preprocess DEM

Preprocessing elevation (DEM) is the first step in the modeling sequence. The DEM defines the cell size and projection for all later processing. Other input files with different projections will be projected to the DEM's projection. Other input rasters will be sampled or aggregated if their input cell size doesn't match the DEM.

Tool input parameters:

1. ***Project geodatabase*:** Navigate to the project geodatabase that you created during the setup procedure.
2. ***Elevation***: Select the elevation raster (DEM) for your study area. If no study area file is entered as a third parameter, the outline of the DEM defines the study area.
3. ***Study area (optional):*** select the study area feature class that defines the boundary of the processing. If the DEM is larger than the study area, the DEM will be masked to the size of the study area.

When the tool completes, three new files will be added to the project: 'dem', 'study_area_polygon', and 'study_area_raster'. This tool loads the project DEM into the Table of Contents pane.

Step 1b (optional): Preprocess Hazard

This tool is available for study areas that don't have a pre-defined safe zone. The safe zone identifies the locations where the evacuation modeling will try to route all locations in the hazard zone. If you just have a hazard zone but no safe zone, use this tool to create a preliminary safe zone. This is created by erasing the hazard zone from the study area. However, this process may also create polygon slivers that need to be removed so modeling won't direct evacuation pathways to these locations. The removal of slivers is a manual process to be done by you after this tool is complete. Once the preliminary safe zone is free of erroneous locations, it can be added to the project using the Preprocess Safezone tool.

Tool input parameters:

1. ***Project geodatabase*:** Navigate to the project geodatabase that you created during the setup procedure.
2. ***Hazard zone:*** Select the hazard zone polygon feature class for your study area.

When the tool completes, two new files named 'hazard' and 'prelim_safe_zone' will be added to the project gdb and 'prelim_safe_zone' will be added to the Table of Contents pane. This preliminary safe zone is now ready for your inspection and editing.

Step 1c: Preprocess Safezone

Use this tool to add a verified safe zone to the project for use by later processing. If you edited and verified the preliminary safe zone from step 1b, just select this file as the input safe zone. If you have a verified safe zone from another source, use this step to enter it into the project data storage. The projection of the input file will be checked against the dem, the input file will be clipped by the study area, and the safe zone will be stored both in vector and raster form.

Tool input parameters:

1. ***Project geodatabase*:** Navigate to the project geodatabase that you created during the setup procedure.
2. ***Safe zone***: Select the verified safe zone polygon feature class for input into the project gdb.

When the tool completes, two new files named 'safe_zone_polygon' and 'safe_zone_raster' will be added to the project data storage. No files are added to the Table of Contents pane.

Step 1d: Preprocess Landcover

This tool preprocesses the various input landcover layers to create the cost inverse raster for use by the Create Evacuation Surfaces tool. Each input layer is merged on top of the previous layer(s) according to processing order, with the base landcover layer as the starting layer. Multiple input files can be combined into a single output in this tool, and a single landcover file such as a roads layer can also be used as the base layer with no other input. This 'roads-only' landcover layer can be useful when travel must be constrained to the road network.

The cost inverse raster is created by setting each landcover type to a Speed Conservation Value (SCV). Speed conservation values represent the fraction of a maximum speed that could be achieved across the given landcover type. Values fall on a spectrum from zero (meaning travel is not possible) to 1.0 (meaning pedestrian evacuation speed is 100 percent of the base travel rate). The tool window will allow you to enter an SCV value for each type of landcover in your input layer(s).

Tool input parameters:

1. ***Project geodatabase*:** Navigate to the project geodatabase that you created during the setup procedure.
2. ***Landcover scenario name*:** Select a landcover name that will be appended to the names of all subsequent files created for this scenario. To limit the length of output file names, it is best to keep the scenario name short. Do not use any special characters except the underscore ("_").
3. ***Base landcover layer*:** Navigate to the base landcover layer for this scenario. This layer can be either a polygon feature class, a polyline feature class, or a raster. If a raster, the data type should be Integer, representing categorical data.
4. ***Landcover field name*:** Select the integer or text attribute name that identifies the landcover type(s) within the file.
5. ***Base layer values*:** add all the unique values from the base layers and assign a SCV value for each landcover type. Figure 2 shows some example values. You can also see these values by hovering over the blue 'i' symbol next to the parameter name in the tool window. Section 3.2.3 of article [1] in References contains descriptions of these values.

LandcoverType	SCV value
Water	0.0
Buildings	0.0
Sand or wetlands	0.5556
Heavy brush	0.6667
Light brush	0.8333
Developed	0.9091
Roads	1.0

Figure 2. Example values for Speed Conservation Values (SCV)

6. ***Additional feature layers**:* For each additional feature class to be added onto the base layer, select the file name, assign a unique processing order value between 1 and 10 to identify the merge order, and enter a SCV value. Additional feature classes can be added by selecting 'Add another'.
7. ***Additional raster layers:*** For each additional raster to be added onto the base layer, select the file name, assign a unique processing order value between 1 and 10 to identify the merge order, and enter a SCV value. Additional rasters can be added by selecting 'Add another'.

Processing order

The base landcover layer is the foundation layer for building the cost inverse raster. Additional layers such as roads, water bodies, fences, and buildings can also be added, and these will be overlaid on the base landcover layer according to the assigned processing order. This value is

used to determine the merge order for the input layers (the base layer is always the first). For example, to create a cost inverse raster with a base layer overlaid by water and then by roads, add the water layer as an additional layer with a processing order of '1' and the roads as an additional layer with a processing order of '2'. This will first overlay the water onto the base layer and then overlay the roads, which will retain the roads (bridges) over water. Table 1 shows this ordering, and Figures 3 and 4 show the landcover layering for full landcover and roads-only processing.

Table 1. Typical landcover layer ordering for the cost inverse raster

ORDER	LANDCOVER LAYER
0 (always the first layer)	Base landcover
1	Water
2	Roads
3	Buildings

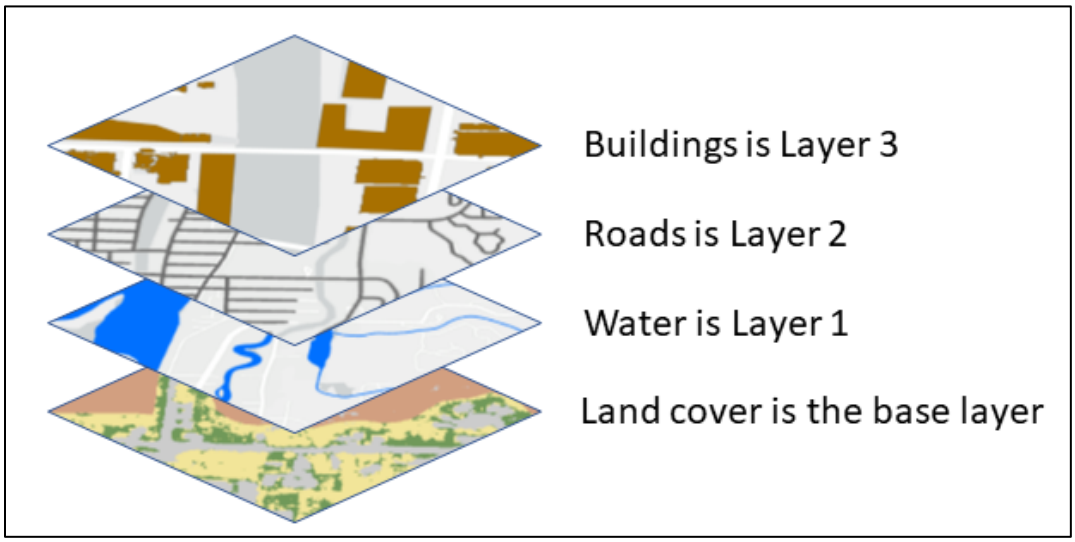


Figure 3. Example layer ordering for a full landcover preprocessing

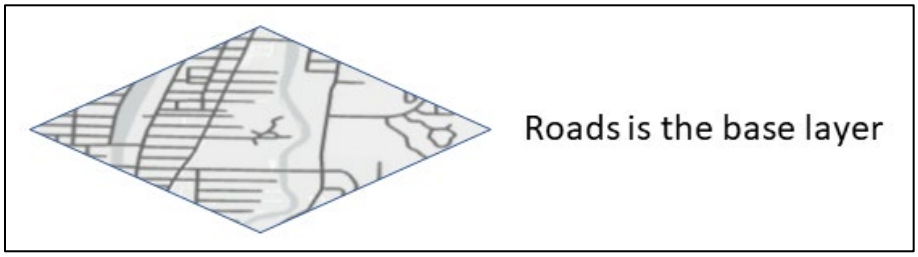


Figure 4. For roads only preprocessing, only the roads layer is used

If you have two additional feature classes and one additional raster dataset, and you want the raster to be merged in between the two feature classes, use the processing order to define this as in Figure 5.

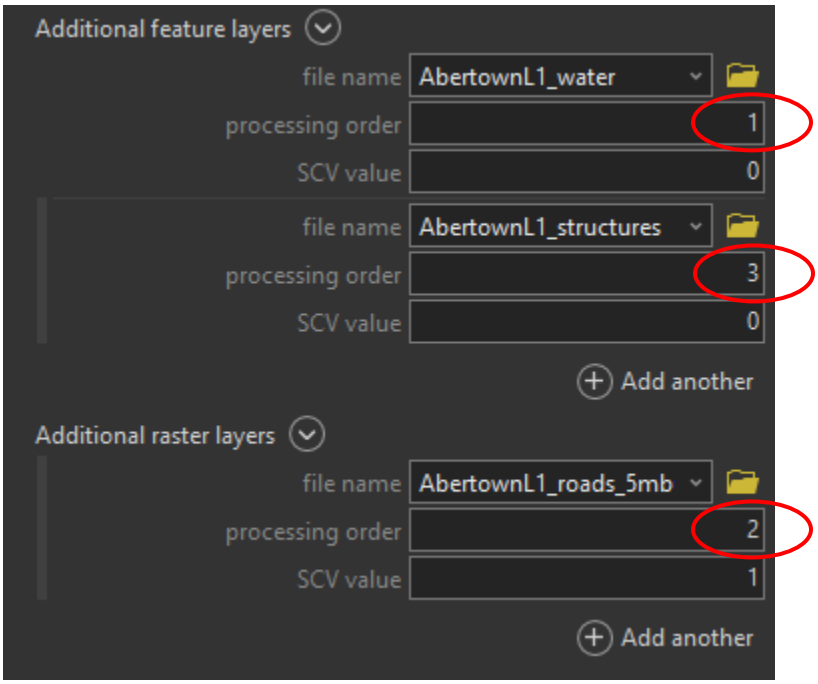


Figure 5. Example selection of processing order when using both feature classes and rasters

When the tool completes, a new file called "lc_cost_inverse_<landcover scenario name>" is created by this step and is added to the Table of Contents pane. Values in the resulting raster are best shown as unique values using standard ArcPro symbology tools. Lower values denote easier travel routes for evacuees. For example, roads entered with a SCV value of 1 will have a cost inverse value of 1.0, while heavy brush entered with a SCV of 0.6667 will have a cost inverse value of 1.5.

Step 2 Time and Speed Maps

These tools are used to create the evacuation modeling output products. They use the elevation, study area, safe zone, and cost inverse layers that are stored in the project data storage, and so the preprocessing must be run before using any of the time and speed map tools. The elevation, study area, and safe zone files are created once for the project while multiple landcover scenarios can be processed for a project. All the time and speed map tools are based on an individual scenario.

Step 2a: Create Evacuation Surface

This tool runs the Spatial Analyst Distance Accumulation tool and creates an evacuation surface showing travel time in minutes. The DEM, safe zone, and cost inverse raster are inputs

to the tool, which calculates the path distance from every cell in the hazard zone to the nearest cell in the safe zone. The travel speed is then used to create a travel surface in minutes.

Tool input parameters:

1. ***Project geodatabase*:** Navigate to the project geodatabase that you created during the setup procedure.
2. ***Landcover scenario name*:** choose the land cover scenario for evacuation modeling.
3. ***Speed value:*** Enter a travel speed value as a decimal number. Each travel speed value added to the list will be used to generate a new evacuation surface. Figure 6 shows some examples of travel speeds. You can also see these values by hovering over the blue 'i' symbol next to the parameter name in the tool window.

speed name	speed value
Impaired walk	0.89
Slow walk	1.2
Fast walk	1.52
Slow run	1.79
Fast run	3.85

Figure 6. Example evacuation travel speeds, in meters per second

When the tool completes, a floating-point raster grid showing pedestrian travel time out of the hazard zone in minutes will be created and added to the Table of Contents pane for each travel speed entered in the tool window. Rasters are named as "evac_surface_<landcover scenario name>_<travel speed>." These rasters may include extraneous travel times because of unrealistic landscape conditions. In the next step, users can cap the maximum time value in a final time map.

Step 2b (optional): Check for Time Outliers

This optional tool is available for evacuation surfaces that include erroneous travel times because of unrealistic landscape conditions, which may have been caused by an abrupt and significant elevation change between one cell in the DEM and the neighboring cells. These unusual travel times will appear in the raster's histogram of values as outliers from the typical range of time values. To remove these outliers, select the evacuation surface and enter a maximum realistic travel time. The evacuation surface will be reclassified to cap all outliers to the maximum travel time entered and the updated surface will be overwritten in the project data storage.

Tool input parameters:

1. ***Project geodatabase*:** Navigate to the project geodatabase that you created during the setup procedure.
2. ***Scenario*:** Choose the land cover scenario name used for creating the evacuation surfaces to be adjusted.
3. ***Maximum times:*** Use this parameter to select any evacuation surface that needs to have unusually large travel times reclassified to some more reasonable limit. Select a surface name and then enter a maximum realistic travel time in minutes.

When the tool completes, the evacuation surfaces selected will be reclassified and updated in the project raster storage. No files are added to the Table of Contents pane by this tool.

Step 2c: Create Time Map

This tool converts the floating-point, evacuation surfaces into integer time maps at 1-minute intervals in both raster and vector form. If building outlines are entered, the time maps are generated with the building footprints filled in with surrounding values.

Tool input parameters:

1. ***Project geodatabase*:** Navigate to the project geodatabase that you created during the setup procedure.
2. ***Landcover scenario name*:** Choose the land cover scenario name used for creating the evacuation surfaces.
3. ***Evacuation surfaces:*** Select one or more evacuation surfaces to process into time maps.
4. ***Buildings for fill (optional):*** Set this optional parameter to a polygon feature class of building footprints if the feature class was used during landcover preprocessing. If added, the building footprints will be filled in with surrounding values in the time map to create a smooth surface. This may aid in your later population processing if your population points are located within building footprints. The footprint holes will be filled in so the Pro Identity tool can be run to attach a travel time value for each population point. Population processing is an optional step for the user and not part of the PEAT processing.

When the tool completes, a time map polygon feature class and a time map raster will be added to the Table of Contents pane for each evacuation surface selected. The feature class is named as “time_map_polygon_<landcover scenario name>_<speed>” and the raster is named as “time_map_raster_<landcover scenario name>_<speed>”. If the maps have buildings filled back in, this will be indicated by ‘fill’ inserted before the scenario name.

Step 2d (optional): Create Vertical Evacuation Map

Vertical-evacuation (VE) structures, such as buildings or berms, are substitutes for naturally occurring high ground within a hazard zone. They are designed to withstand the

sudden-onset hazard and provide a safe haven for nearby population evacuation. Multiple potential VE sites can be analyzed to determine the populations served by each structure, and this information can aid in the site selection decision-making process [4].

This tool creates vertical evacuation maps from a base time map and input polygons of VE locations. The tool takes the selected base time map and goes through all the input VE polygons one at a time, merging each VE polygon with the original safe zone and creating new evacuation surfaces and time maps. Time maps are created for each individual VE polygon that is contained in the user-supplied polygon layer. Developing a travel-time map that recognizes all VE polygons simultaneously requires a VE layer where multiple polygons are merged into a single multi-part polygon and provided as input to the tool.

Tool input parameters:

1. ***Project geodatabase*:** Navigate to the project geodatabase that you created during the setup procedure.
2. ***Landcover scenario name*:** Choose the land cover scenario name used for creating the vertical evacuation time maps.
3. ***Base time map:*** Choose an existing time map to define the travel speed for the vertical evacuation time maps.
4. ***VE features:*** Select the polygon feature class containing the locations of potential evacuation structures.
5. ***VE number field:*** Select an attribute field that contains unique integer values identifying each VE polygon.

When the tool completes, the VE travel time maps in both vector and raster form are added to the Table of Contents pane with the same name as the base time map and "VE<number>" appended after the scenario name. The VE number is taken from the VE number field entered.

Step 2e (optional): Create Speed Map

This tool creates speed maps showing the locations where successful evacuation is likely at a given travel speed based on multiple time maps and factoring in event arrival time and travel delay [10]. The user supplies an estimated event-arrival time (in minutes) and can optionally add an evacuation-delay estimate (in minutes). The evacuation delay factors in the time it might take people to start the evacuation after signs such as earthquake shaking or community warning systems alert them to the approaching hazard.

Tool input parameters:

1. ***Project geodatabase*:** Navigate to the project geodatabase that you created during the setup procedure.

2. ***Landcover scenario name*:** Choose the land cover scenario name used for creating the time maps to be combined for the speed map.
3. ***Time maps:*** Select the time maps to use in the creation of the speed map.
4. ***Event arrival time in minutes:*** Set this parameter to the time in minutes when the hazard event is estimated to arrive in the study area.
5. ***Delay in minutes (optional):*** Set this parameter to the number of minutes it is estimated individuals will take before beginning to evacuate.

When the tool completes, the speed map will be added to the Table of Contents pane in both vector and raster form with the name "speed_map_<raster or vector>_<land cover scenario name>_<arrival time>_<departure delay>."

Step 3 Basins

Step 3 (optional): Create Evacuation Basins

This tool performs a roads-only analysis on a study area to identify likely pathways for evacuees to the nearest safe zone [12]. This information could help practitioners identify evacuation corridors and, if population data is added, determine which routes are likely to be heavily used. This tool also identifies evacuation basins, which outline neighborhoods that share common evacuation pathways to safety. This information can help guide individuals to safety in urban areas where optimal routes may be difficult to discern. Along with pathways and evacuation basins, this tool outputs pour points, which are point locations identifying where the flow accumulation lines meet the safe zone. These points contain the counts of population expected to be arriving at each point.

Tool input parameters:

1. ***Project geodatabase*:** Navigate to the project geodatabase that you created during the setup procedure.
2. ***Roadlines*:** Navigate to a polyline feature class that contains the roads for the study area. This feature class does not need to be in a network.
3. ***Population points (optional):*** Set this parameter to a point feature class of population locations.
4. ***Population count (optional):*** Set this parameter to a column name in the attribute table of the population file that contains population counts for each point (this column can be either integer or decimal).

When the tool completes, the pour points ("pourpoints_points_PEATbasin"), flow lines ("flow_accumulation_lines_PEATbasin"), and watershed ("watersheds_bounding_polygon_PEATbasin") files will be loaded into the table of contents. The watersheds bounding polygons were created by determining the minimum bounding geometry around each watershed for better visualization.

Utilities

Delete Scenario

Use this tool to remove all the files, both feature class and raster, that belong to a particular scenario, including the Basin scenario. The tool will delete all files, folders, and geodatabase datasets associated with the selected scenario.

Tool input parameters:

1. ***Project geodatabase*:** Navigate to the project geodatabase that you created during the setup procedure.
2. ***Scenario:*** Select the scenario to be deleted.

USGS Disclaimer

Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

File Naming Convention

The feature classes and rasters created during the PEAT processing are named according to a convention that attaches information such as the file type, scenario name, and related travel speed or VE number. This naming convention is designed to make it easier to find and interpret the files in the ArcPro Table of Contents and the storage structure.

- Preprocess Landcover
  - lc_cost_inverse_< scenario name>
- Create Evacuation Surface
  - evac_surface_< scenario name>_<travel speed>
- Create Time Map
  - time_map_raster_<scenario name>_< travel speed>
  - time_map_raster_filled_<scenario name>_< travel speed> (optional)
  - time_map_polygon_<scenario name>_< travel speed>
  - time_map_polygon_filled_<scenario name>_< travel speed> (optional)
- Create Vertical Evacuation Map
  - time_map_raster_<scenario name>_<VE num>_< travel speed>
  - time_map_polygon_<scenario name>_<VE num>_< travel speed>
- Create Speed Map
  - speed_map_raster_<scenario name>_<arrival time>_<delay>
  - speed_map_polygon_<scenario name>_<arrival time>_<delay>
- Create Evacuation Basins
  - flow_accumulation_lines_PEATbasin
  - pourpoints_points_PEATbasin
  - watersheds_bounding_polygon_PEATbasin

Key

- <scenario name> - the name entered for the scenario in the Preprocess Landcover tool
- <travel speed> - the travel speed entered in the Create Evacuation Surface tool, a speed of 1.2 will appear in the name as 1p2
- <arrival time> - the arrival time in minutes entered in the Create Speed Map tool
- <delay> - the optional delay time in minutes entered in the Create Speed Map tool
- <VE num> - the unique integer appended to ‘VE’ for each location from the polygon feature class entered in the Create Vertical Evacuation Map tool

Feature Class Storage map for Project Geodatabase

The project geodatabase holds the feature classes used during PEAT processing. In cases where scenario processing generates multiple files with similar names, the files are grouped within a geodatabase dataset with the scenario name and the processing name. The basin processing is considered as a scenario, and all files for basin processing are contained in the PEATbasin dataset. Figure 7 shows an example project geodatabase structure.

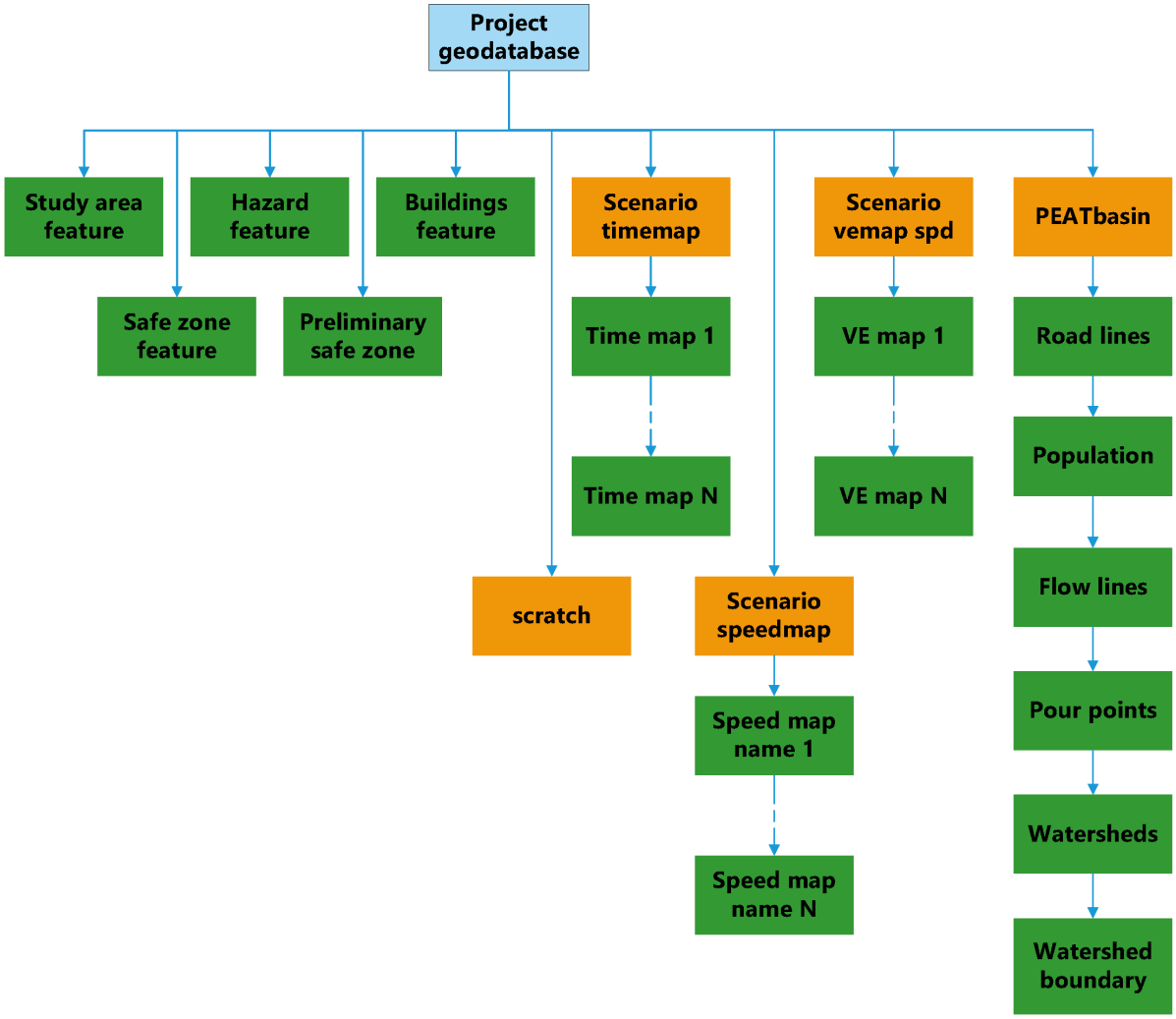


Figure 7. Project geodatabase file structure

Raster Storage Map for Project

The project folder holds the rasters used during PEAT processing. Subfolders within this folder are used to group files by scenario and file type. Figure 8 shows an example folder structure.

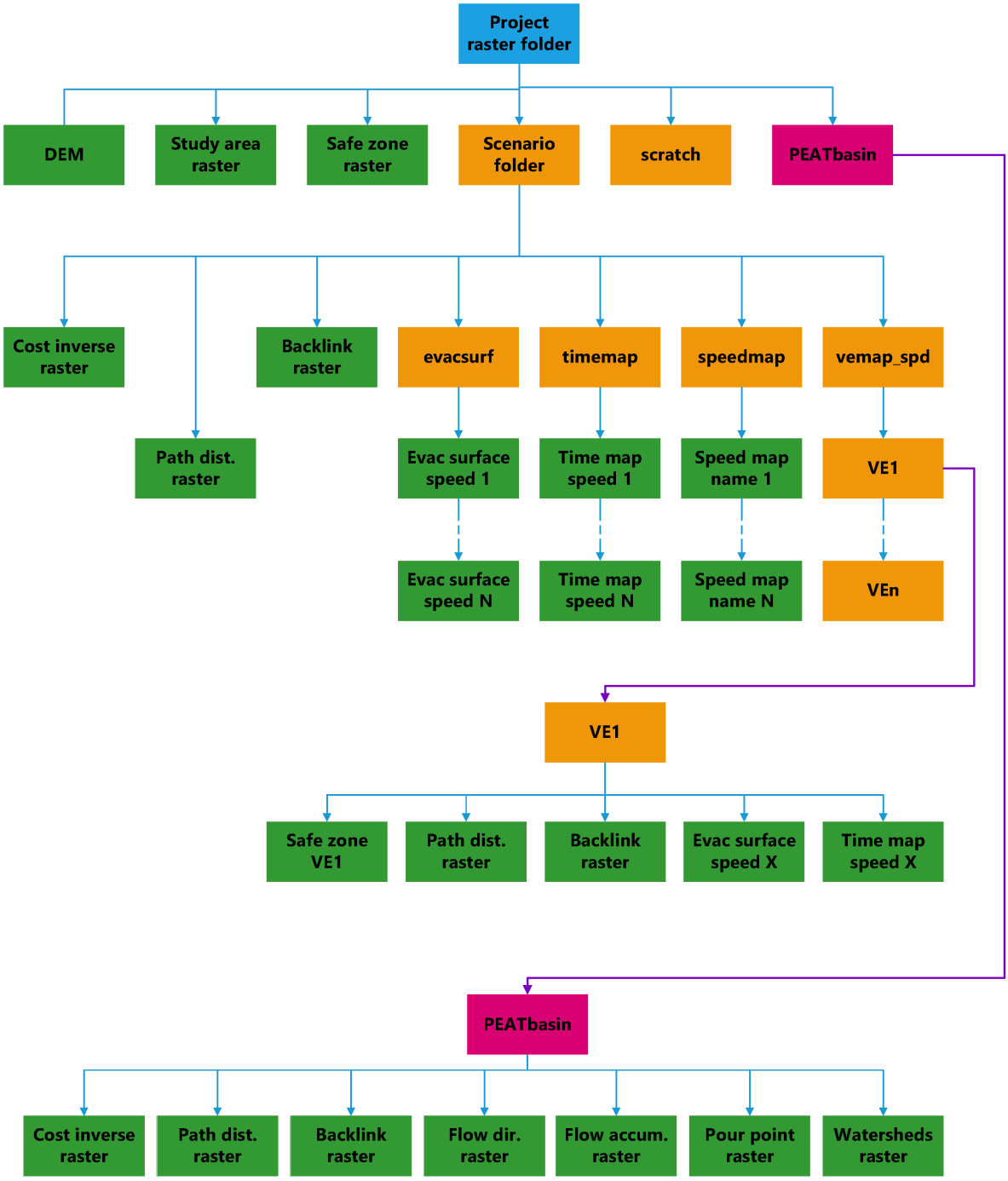


Figure 8. Project folder structure for rasters

References

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